

MICROCOMPUTERS

Users' Manual
Memory Expansion Module
KIM-3B

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The information in this manual has been reviewed and is believed to be entirely reliable. However, no responsibility is assumed for inaccuracies. The material in this manual is for informational purposes only and is subject to change without notice.

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MOS TECHNOLOGY, INC.
950 Rittenhouse Road
Norristown, PA. 19401

CHAPTER 1
Introduction

Congratulations on your purchase of a KIM memory expansion board. It has been carefully engineered to provide high reliability and a long service life. Please make sure you take a few minutes and read this User's Manual completely. You will then be familiar with all the features of your memory expansion board and will find it easy to connect the board to your existing KIM-1 system.

A single memory expansion board may be wired directly to your KIM-1. By using the KIM-4 motherboard you may add additional memory modules to expand your memory space by an additional 58,000 bytes

Like all MOS Technology, Inc. microprocessor modules, your memory expansion module is completely assembled and tested. Even if you are not using a KIM-4 motherboard, all you will have to do is wire a simple cable. Your KIM memory expansion module is covered by a complete 90-day warranty and, like all KIM modules, factory repair services are available even after the expiration of your warranty.

Your KIM memory expansion board has its own +5v regulator and requires only 8 to 10 VDC unregulated for satisfactory operation. If you already have a regulated +5v supply, you may use it with your expansion board and bypass the regulation circuitry.

All the necessary circuitry has been included to make your memory expansion module completely compatible with KIM-1. By setting the switches on the

memory expansion board, you can select the address locations in memory where you wish your expansion memory to reside.

The integrated circuit memories used on your board are high-speed static memory modules. No refresh cycles are required and access to this memory will not require any slow-down of your KIM-1.

Chapter 2 of this manual explains how to install your new memory expansion module in your KIM system. Chapter 3 explains how to check out your memory expansion module and how to test it if you ever suspect that it has failed. Chapter 4 contains information on your memory expansion module warranty, and Chapter 5 explains the theory of operation. If for any reason you are unable to get your memory expansion module operating satisfactorily, follow carefully the checkout instructions in Chapter 3. If you are still unable to get satisfactory operation, return the module as described in Chapter 4 or contact the manager of KIM Customer Support at MOS Technology, Inc. corporate headquarters, 950 Rittenhouse Road, Norristown, PA 19401.

CHAPTER 2

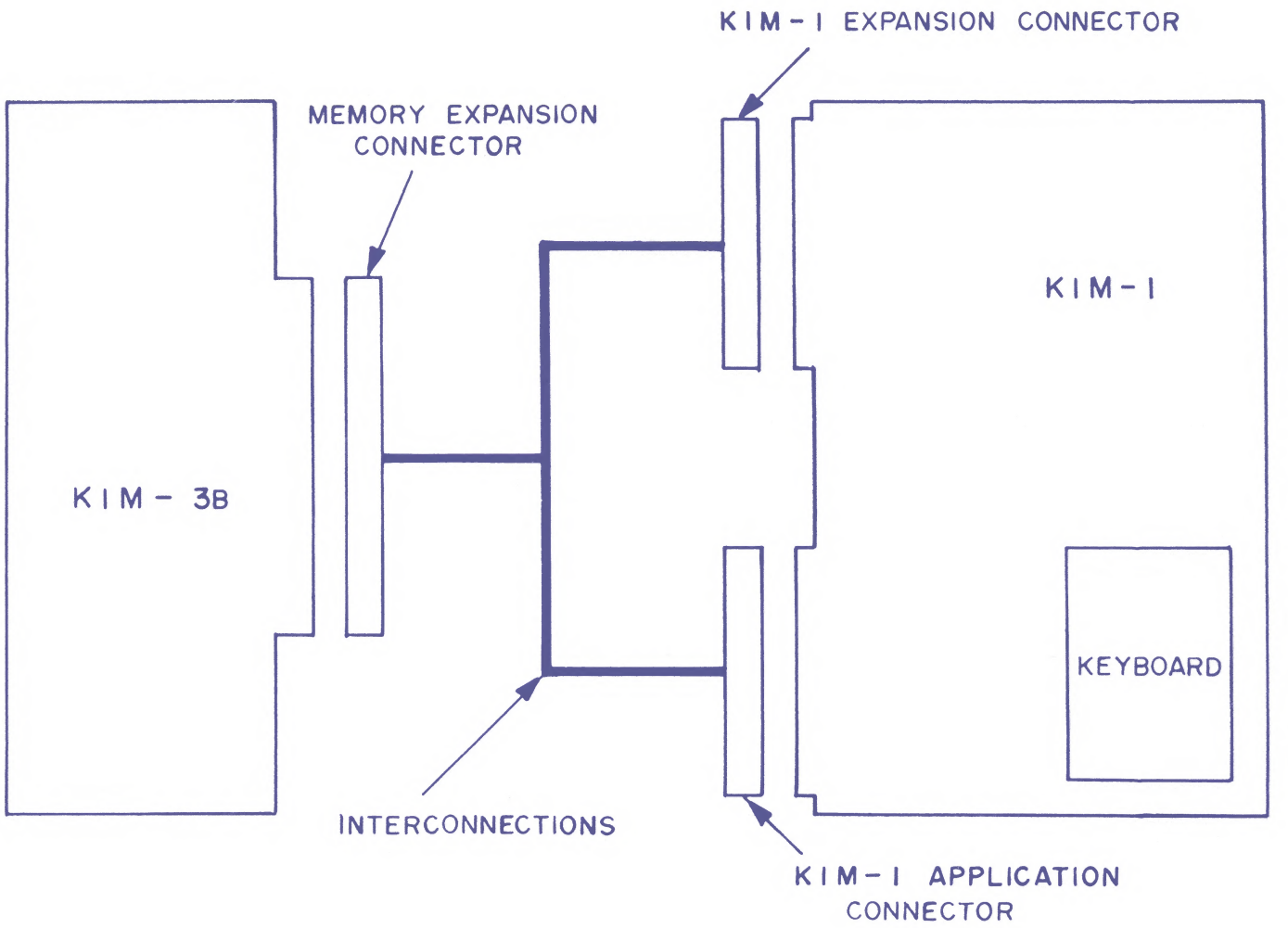
Installation

2.1 INTRODUCTION

How you install your KIM memory expansion module will depend on whether or not you are using a KIM-4 motherboard to interface your expansion module to KIM-1. The pin configuration on your KIM-3B allows you to plug your memory expansion module directly into the KIM-4 motherboard and begin operation immediately. You may insert as many KIM-3B memory expansion modules into the motherboard as you wish, taking care that each is set to a different memory location. If, however, you are connecting your memory expansion module directly to KIM-1, only one KIM-3B may be connected in this manner. In this case you will have to wire a cable for your memory expansion module which connects to the KIM-1 expansion and application connectors. (See Figure 1)

2.2 CONNECTING YOUR MEMORY EXPANSION MODULE TO THE MOTHERBOARD (KIM-4)

If you have already installed a KIM-4 motherboard in your KIM system, it is only necessary to plug the KIM-3B into any slot on the motherboard. Make sure that your motherboard power supply has sufficient capacity to supply the needs of your memory expansion module. See your KIM-4 User Manual for power supply connections. The KIM-3B memory expansion module draws 2.1A maximum. Your KIM memory expansion module should be inserted in the motherboard so that the component side of the board faces away from the end of the motherboard to which KIM-1 is connected. Prior to inserting the memory board in the motherboard set the address switches located on the memory board to the correct position


KIM-1 EXPANSION CONNECTOR
MEMORY EXPANSION CONNECTOR
KIM-1
KIM-3B
KEYBOARD
INTERCONNECTIONS
KIM-1 APPLICATION CONNECTOR

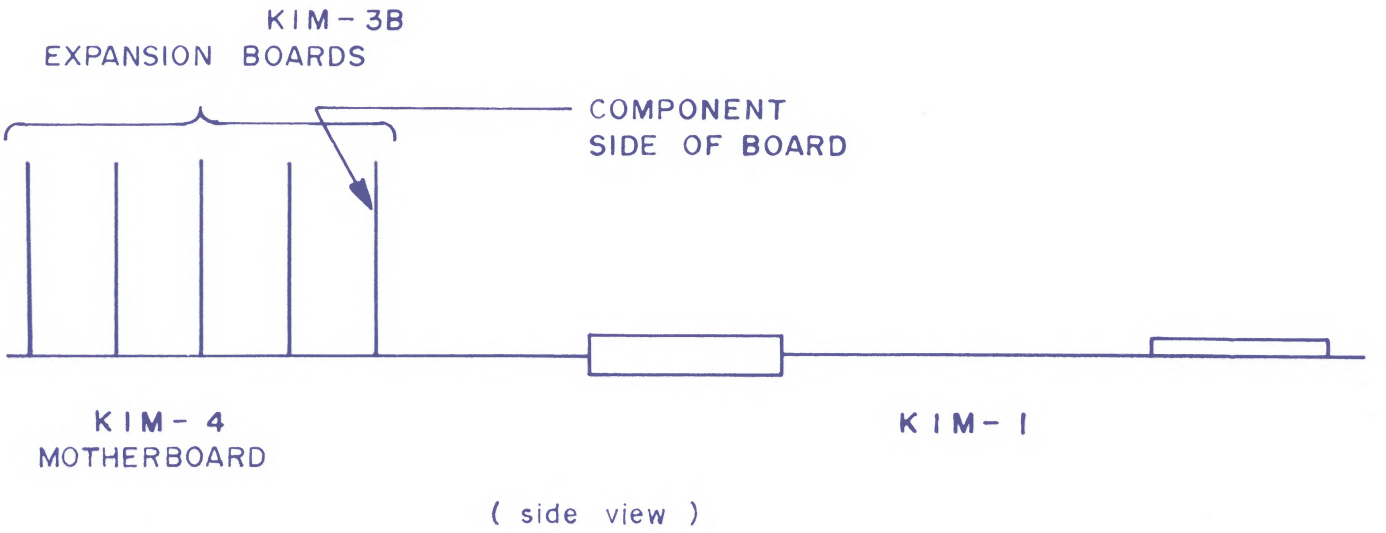

KIM-3B EXPANSION BOARDS
COMPONENT SIDE OF BOARD
KIM-4 MOTHERBOARD
KIM-1
(side view)

Figure 1

for the address in KIM's memory space where you wish to have the additional memory reside. Setting the address switches is described below. When the memory switches are set and the card is inserted in the motherboard you are ready to check out the operation of the board. See Chapter 3 for this operation.

2.3 CONNECTING YOUR MEMORY EXPANSION MODULE WITHOUT A MOTHERBOARD

If you are connecting your memory expansion module directly to a KIM-1, it will be necessary for you to wire a cable to connect the memory expansion module connector to the application and expansion connectors on your KIM-1. Note that if you wish to use the same +5v supply which presently powers your KIM-1, that supply should be connected to pin 21 and pin Y on your memory expansion module connector. If, however, you wish to use an unregulated +8v to +10v supply, that unregulated voltage should be connected to pins 19 and 20 on your memory expansion module. Unregulated +8v to +10v should never be connected to your KIM-1. In either case, insure that your supply can provide at least 1.3A for KIM-3B.

Tables 2A and 2B show the interconnection between KIM-1 and your memory expansion module. If you do not have the appropriate connectors for the KIM-1 expansion connector and your KIM-3B memory expansion module, they can be obtained from most electronic parts supply houses. They are manufactured by Vector and their part number is R644. Note that the pin designation is marked next to each pin on the connector. Once you have wired the cable interconnecting the three connectors, carefully recheck your wiring for incorrectly placed wires or inadvertent short circuits. Note that the wiring table shows the pins on the KIM-1 connectors are preceded by an "A" or "E." The "A" indicates that the connection should be made to the appropriate pin on the applications connector; the "E" indicates that the pin is on the expansion connector.

TABLE 2A

KIM-3B Connector Pin	KIM-1 Connections
1	A-1
2	No Connection
3	No Connection
4	No Connection
5	No Connection
6	No Connection
7	No Connection
8	E-8
9	E-9
10	E-10
11	E-11
12	E-12
13	E-13
14	E-14
15	E-15
16	A-K (Remove Jumper from A-1)
17	No Connection
18	No Connection
19	+8v } Connect only +5v or +8v, NOT BOTH
20	+8v }
21	+5v }
22	A-1

NOTE: Cable from KIM-1 to KIM-3B must be six inches in length or less.

TABLE 2B

KIM-3B Connector Pins	KIM-1 Connections
A	A-1
B	E-A
C	E-B
D	E-C
E	E-D
F	E-E
H	E-F
J	E-H
K	E-J
L	E-K
M	E-L
N	E-M
P	E-N
R	E-P
S	E-R
T	E-S
U	E-T
V	E-U
W	E-V
X	E-Y
Y	To +5v (If +8v NOT USED)
Z	A-1

2.4 SETTING THE ADDRESS SWITCHES

In order to make your memory expansion module as versatile as possible, we have included four switches to allow you to place your expansion memory at any memory address (see Figure 3). We suggest that you place your first memory expansion module starting at address 2000_{hex} and continue to expand your memory into successively higher memory locations. Tables 4 and 5 indicate the switch settings for various memory locations using your KIM-3B. Be sure to consult the appropriate table for the module you have purchased. Once you have chosen the memory space for your expansion module and correctly set the addressing switches, turn off all power and insert the memory module in its connector. You are now ready to test your memory expansion module.

Note: Do not set the switches so that your expansion memory has an address below 2000_{hex} as it will conflict with the memory and other circuitry in your KIM-1. It is not possible to put your expansion memory in the Memory block $0400\text{-}1400_{hex}$ already decoded on KIM-1.

KIM-3B Address Switch Setting

When Address Switch Is: 3 2 1	Lowest Address Is:	Highest Address Is:
0 0 0	0000	1FFF (do not use)
0 0 X	2000	3FFF
0 X 0	4000	5FFF
0 X X	6000	7FFF
X 0 0	8000	9FFF
X 0 X	A000	BFFF
X X 0	C000	DFFF
X X X	E000	FFFF

X = Switch is NOT on

0 = Switch IS on

Note: Switch 4 will logically disconnect the board when closed. Switch 5 is used for write protection when left open.

CHAPTER 3

Checkout and Test Program

3.1 Your memory expansion module has been carefully tested to assure correct operation. In this section we will describe how you can briefly check the operation of your memory expansion module. We have also included a test program which will allow you to verify correct operation of all memory cells in your memory expansion module. It should only be necessary to run this program if you suspect that the memory module has failed.

To verify that your memory expansion module has been correctly wired and that the address switches are correctly set, just address some of the memory locations and verify that you can change the contents of those locations. Using the keypad provided with your KIM-1, and assuming that you have set the address switches on your memory expansion module so that the lowest address is 2000, use the following procedure:

Checkout and Test Program

Depress Key	See Displayed
/RS/	XXXX XX
/AD/	XXXX XX
/2/ /0/ /0/ /0/	2000 XX
/DA/	2000 XX
/3/ /A/	2000 3A
/7/ /9/	2000 79
/+/	2001 XX
/3/ /7/	2001 37

If you are unable to change the data in memory there are two possible sources of trouble:

1. The memory expansion board is not correctly connected to KIM-1. If you are using a motherboard, check that the motherboard is correctly installed and that the memory expansion card has the component side of the board facing away from the KIM-1. If you are not using a motherboard, carefully check your wiring against the list provided in Tables 2A and 2B.
2. You have incorrectly set the memory address switches. The figure below shows the appearance of the memory address switch when it is configured so that the lowest expansion memory address is 2000_{16}. Recheck the information in Chapter 2 if you are unsure of the placement of the memory expansion module.

3.2 TEST PROGRAM FOR MEMORY CHECKOUT

Although your KIM memory expansion module has been carefully tested before shipment, like any other electronic device, it can fail in use. If you suspect that your memory expansion module is not working correctly, the following program can be used to check the memory operation. It should be noted that programs for testing memory modules for all possible failure modes would be quite complex and require lengthy running time on KIM. The following programs simply write and read every possible bit pattern in every memory location. They do not check, for instance, whether writing to a given memory location may also affect other memory locations.

3.3 CHECKING YOUR MEMORY FROM A TERMINAL

The following program assumes that you have a terminal connected to the serial input and output ports of your KIM-1. To use the program, type it into KIM memory starting at location 200_{16}; make a paper tape copy once you have loaded the program using the KIM dump routine. To operate the program, load the lowest address which you wish to test in location 0000 and 0001, then load the highest memory address you wish to test in location 0002 and 0003. For instance, to check all memory locations between 2000 and 2FFF you would load 00 in location 0000, 20 in location 0001, FF in location 0002, and 2F in location 0003. To operate the program, load address 022A and hit the G key. The program will then fill the specified memory locations with 0's and then read all locations to verify that the zero has been written. It will then load the specified memory with 01 and again verify the data. The process will continue until all bit patterns from 00 to FF have been written and read correctly. If any memory location fails to read or write correctly the address of the defective cell will be written to the terminal, along with the code which would not

Program 1

```
CARD = LOC      CODE        CARD
     1                      BOTLO=$0
     2                      BOTHI=$1
     3                      TOPLO=$2
     4                      TOPHI=$3
     5                      PTRLO=$4
     6                      PTRHI=$5
     7                      MASK=$6
     8                      PRTBYT=$1E3B
     9                      OUTSP=$1E9E
    10                      CRLF=$1E2F
    11  0000                *=$200
    12  0200  E6 04         INCPTR  INC PTRLO
    13  0202  D0 02                 BNE END
    14  0204  E6 05                 INC PTRHI
    15  0206  60            END     RTS
    16
    17                        ERROR ROUTINE FOR TTY
    18
    19  0207  A5 05         ERROR   LDA PTRHI
    20  0209  20 3B 1E              JSR PRTBYT
    21  020C  A5 04                 LDA PTRLO
    22  020E  20 3B 1E              JSR PRTBYT
    23  0211  20 9E 1E              JSR OUTSP
    24  0214  A5 06                 LDA MASK
    25  0216  20 3B 1E              JSR PRTBYT
    26  0219  20 2F 1E              JSR CRLF
    27  021C  60                    RTS
    28
    29                         INITIALIZATION SUBROUTINE
    30
    31  021D  A9 00         INIT LDA =$00
    32  021F  85 06                 STA MASK
    33  0221  A5 00         RESET    LDA BOTLO
    34  0223  85 04                 STA PTRLO
    35  0225  A5 01                 LDA BOTHI
    36  0227  85 05                 STA PTRHI
    37  0229  60                    RTS
    38
    39                         MAINLINE FOR MEMORY TEST
    40
    41  022A  D8            BEGIN CLD
    42  022B  20 1D 02              JSR INIT
    43  022E  A0 00                 LP1 2DY #$00
    44  0230  A5 06         WRLOOP  LDA MASK
    45  0232  91 04                 STA (PTRLO),Y
    46  0234  20 00 02              JSR INCPTR
    47  0237  A5 04                 LDA PTRLO
    48  0239  C5 02                 CMP TOPLO
    49  023B  D0 F3                 BNE WRLOOP
    50  023D  A5 05                 LDA PTRHI
    51  023F  C5 03                 CMP TOPHI
    52  0241  D0 ED                 BNE WRLOOP
    53  0243  20 21 02              JSR RESET
    54  0246  B1 04         RDLOOP  LDA (PTRLO),Y
    55  0248  C5 06                 CMP MASK
    56  024A  F0 03                 BEQ CONT
    57  024C  20 07 02              JSR ERROR
    58  024F  20 00 02      CONT    JSR INCPTR
    59  0252  A5 04                 LDA PTRLO
    60  0254  C5 02                 CMP TOPLO
    61  0256  D0 EE                 BNE RDLOOP
    62  0258  A5 05                 LDA PTRHI
    63  025A  C5 03                 CMP TOPHI
    64  025C  D0 E8                 BNE RDLOOP
    65  025E  E6 06                 INC MASK
    66  0260  D0 03                 BNE CYCLE
    67  0262  20 2F 1E              JSR CRLF
    68  0265  20 21 02      CYCLE    JSR RESET
    69  0268  4C 2E 02              JMP LP1

END OF MOS/TECHNOLOGY 650X ASSEMBLY VERSION 4
NUMBER OF ERRORS =    0,  NUMBER OF WARNINGS =    0
```

read or write correctly. When all bit patterns have been tested in all specified cells, the program will output a carriage return and line feed and begin the entire cycle over again. For a 4K memory expansion module the entire test will take about 1-1/2 minutes.

3.4 CHECKING MEMORY OPERATION WITH THE KEYPAD

Program 2 tests memory in a similar fashion, but does not require a terminal. As in the first program, the address of the lower limit and upper limit of the memory to be checked is inserted in locations 0 through 3. When the program has been keyed in, you will probably wish to record it on your audio cassette for future use. When the starting address (022E) is loaded and the GO button is depressed the program will check memory as described above. However, if a defective cell is encountered the address of the defective cell will be displayed on the leftmost four digits of the display and the program will halt. Pushing any button on the keypad will resume the testing operation. When all memory cells have been checked, a value of 0000 will appear in the display and the program will halt.

Program 2

```
CARD =  LOC      CODE          CARD
     1                       BOTLO=$0
     2                       BOTHI=$1
     3                       TOPLO=$2
     4                       TOPHI=$3
     5                       PTRLO=$4
     6                       PTRHI=$5
     7                       MASK=$6
     8                       POINTH=$FB
     9                       POINTL=$FA
    10                       SCANS=$1F1F
    11                       AK=$1EFE
    12                       RESVEC=$1C22
    13   0000                *=$200
    14   0200   E6 04       INCPTR  INC PTRLO
    15   0202   D0 02               BNE END
    16   0204   E6 05               INC PTRHI
    17   0206   60          END     RTS
    18
    19                        ERROR ROUTINE FOR KEYPAD
    20
    21   0207   A5 05       ERROR   LDA PTRHI
    22   0209   85 FB               STA POINTH
    23   020B   A5 04               LDA PTRLO
    24   020D   85 FA               STA POINTL
    25   020F   A5 06               LDA MASK
    26   0211   85 F9              STA $F9
    27   0213   20 1F 1F   ERI      JSR SCANS
    28   0216   20 FE 1E            JSR AK
    29   0219   F0 F8               BEQ ERI
    30   021B   20 FE 1E   ERLOOP  JSR AK
    31   021E   D0 FB               BNE ERLOOP
    32   0220   60                  RTS
    33
    34                        INITIALIZATION SUBROUTINE
    35
    36   0221   A9 00       INIT LDA =$00
    37   0223   85 06               STA MASK
    38   0225   A5 00       RESET LDA BOTLO
    39   0227   85 04               STA PTRLO
    40   0229   A5 01               LDA BOTHI
    41   022B   85 05               STA PTRHI
    42   022D   60                  RTS
    43
    44                        MAINLINE FOR MEMORY TEST
    45
    46   022E   D8          BEGIN CLD
    47   022F   A0 00               LDY =$00
    48   0231   20 21 02            JSR INIT
    49   0234   A5 06       WRLOOP LDA MASK
    50   0236   91 04               STA (PTRLO),Y
    51   0238   20 00 02            JSR INCPTR
    52   023B   A5 04               LDA PTRLO
    53   023D   C5 02               CMP TOPLO
    54   023F   D0 F3               BNE WRLOOP
    55   0241   A5 05               LDA PTRHI
    56   0243   C5 03               CMP TOPHI
    57   0245   D0 ED               BNE WRLOOP
    58   0247   20 25 02            JSR RESET
    59   024A   B1 04       RDLOOP LDA (PTRLO),Y
    60   024C   C5 06               CMP MASK
    61   024E   F0 03               BEQ CONT
    62   0250   20 07 02            JSR ERROR
    63   0253   20 00 02   CONT     JSR INCPTR
    64   0256   A5 04               LDA PTRLO
    65   0258   C5 02               CMP TOPLO
    66   025A   D0 EE               BNE RDLOOP
    67   025C   A5 05               LDA PTRHI
    68   025E   C5 03               CMP TOPHI
    69   0260   D0 E8               BNE RDLOOP
    70   0262   E6 06               INC MASK
    71   0264   D0 09               BNE LOOP
    72   0266   A9 00               LDA =$00
    73   0268   85 FA               STA $FA
    74   026A   85 FB               STA $FB
    75   026C   4C 22 1C            JMP RESVEC
    76   026F   20 25 02            LOOP JSR RESET
    77   0272   4C 34 02                 JMP WR LOOP
    78                             .END
```

CHAPTER 4
Warranty and Service

Should you experience difficulty with your KIM-3B module and be unable to diagnose or correct the problem, you may return the unit to MOS Technology, Inc. for repair.

4.1 IN-WARRANTY SERVICE

All KIM series microcomputer modules are warranted by MOS Technology, Inc. against defects in workmanship and materials for a period of ninety (90) days from date of delivery. During the warranty period, MOS Technology, Inc. will repair or, at its option, replace at no charge components that prove to be defective provided that the module is returned, shipping prepaid, to:

KIM Customer Service Department
901 California Ave.
Palo Alto, CA 94304

This warranty does not apply if the module has been damaged by accident or misuse, or as a result of repairs or modifications made by other than authorized personnel at the above captioned service facility.

No other warranty is expressed or implied. MOS Technology, Inc. is not liable for consequential damages.

4.2 OUT-OF-WARRANTY SERVICE

Beyond the ninety (90) day warranty period, KIM modules will be repaired for a reasonable service fee. All service work performed by MOS Technology,

Inc. beyond the warranty period is warranted for an additional ninety (90) day period after shipment of the repaired module.

4.3 POLICY OF CHANGES

All KIM series modules are sold on the basis of descriptive specifications in effect at the time of sale. MOS Technology, Inc. shall have no obligation to modify or update products once sold. MOS Technology, Inc. reserves the right to make periodic changes or improvements to any KIM series module.

4.4 SHIPPING INSTRUCTIONS

It is the customer's responsibility to return the KIM series module with shipping charges prepaid to the above captioned service facility.

For in-warranty service, the KIM module will be returned to the customer, shipping prepaid, by the fastest economical carrier.

For out-of-warranty service, the customer will pay for shipping charges both ways. The repaired KIM module will be returned to the customer C.O.D. unless the repairs and shipping charges are prepaid by the customer.

Please be certain that your KIM module is safely packaged when returning it to the above captioned service facility.

CHAPTER 5

Theory of Operation

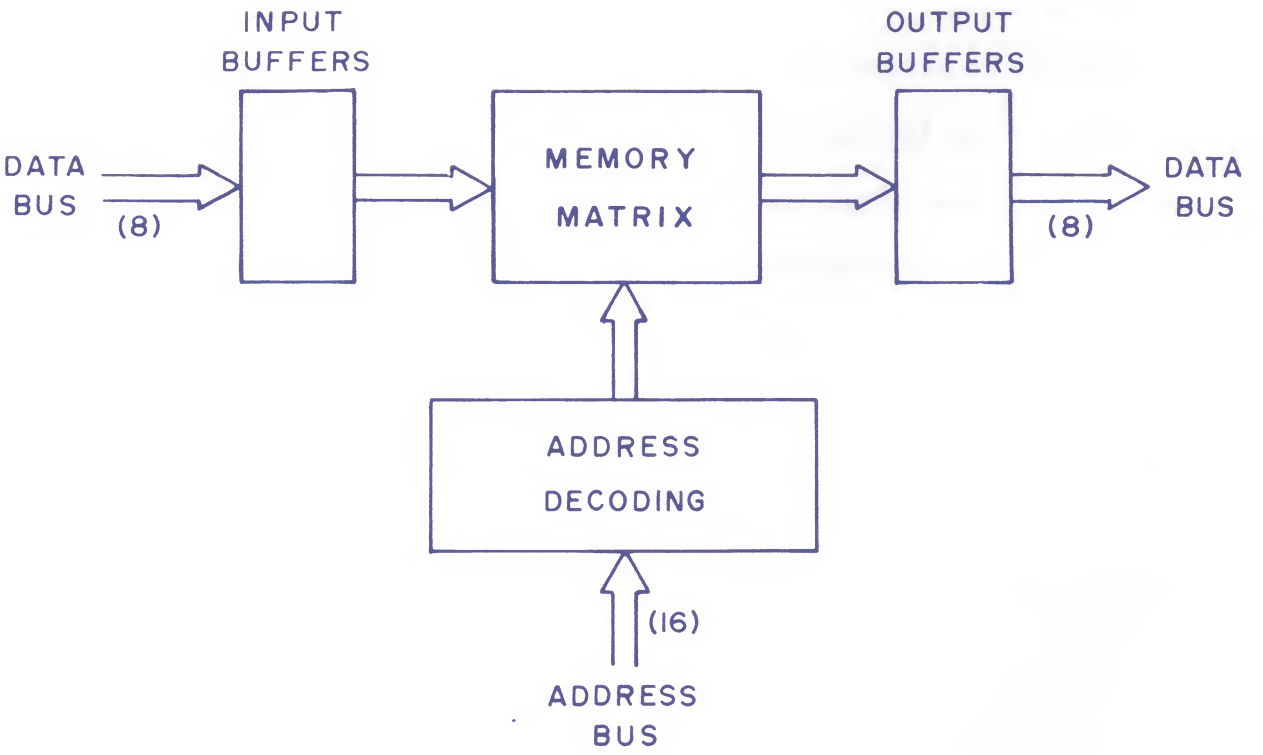
5.1 The schematic shows the interconnection of the components on the KIM-3B board. The diagram below illustrates the pin connections to the 2114-type memories used on the boards. When $\overline{CSX}$ is low, four of the 1024 bits in the package are selected. If $\overline{WE}$ is low, the selected cells will have the values of the I/O lines (1 or 0) written into them. If pin $\overline{WE}$ is high, the contents of the addressed cells will be placed on the I10 lines.

6550

A6	1		18	VCC
A5	2		17	A7
A4	3		16	A8
A3	4	2	15	A9
AO	5	1	14	I/01
Al	6	1	13	I/02
A2	7	4	12	I/03
$\overline{CS}$	8		11	I/04
GND	9		10	$\overline{WE}$

The board is composed of the memory circuits, addressing circuitry, and buffers. An on-board voltage regulator is also provided.

In operation, address bus lines 0 through 9 are buffered by A10, B10 and B11, and connected directly to the memory circuits. The high-order address

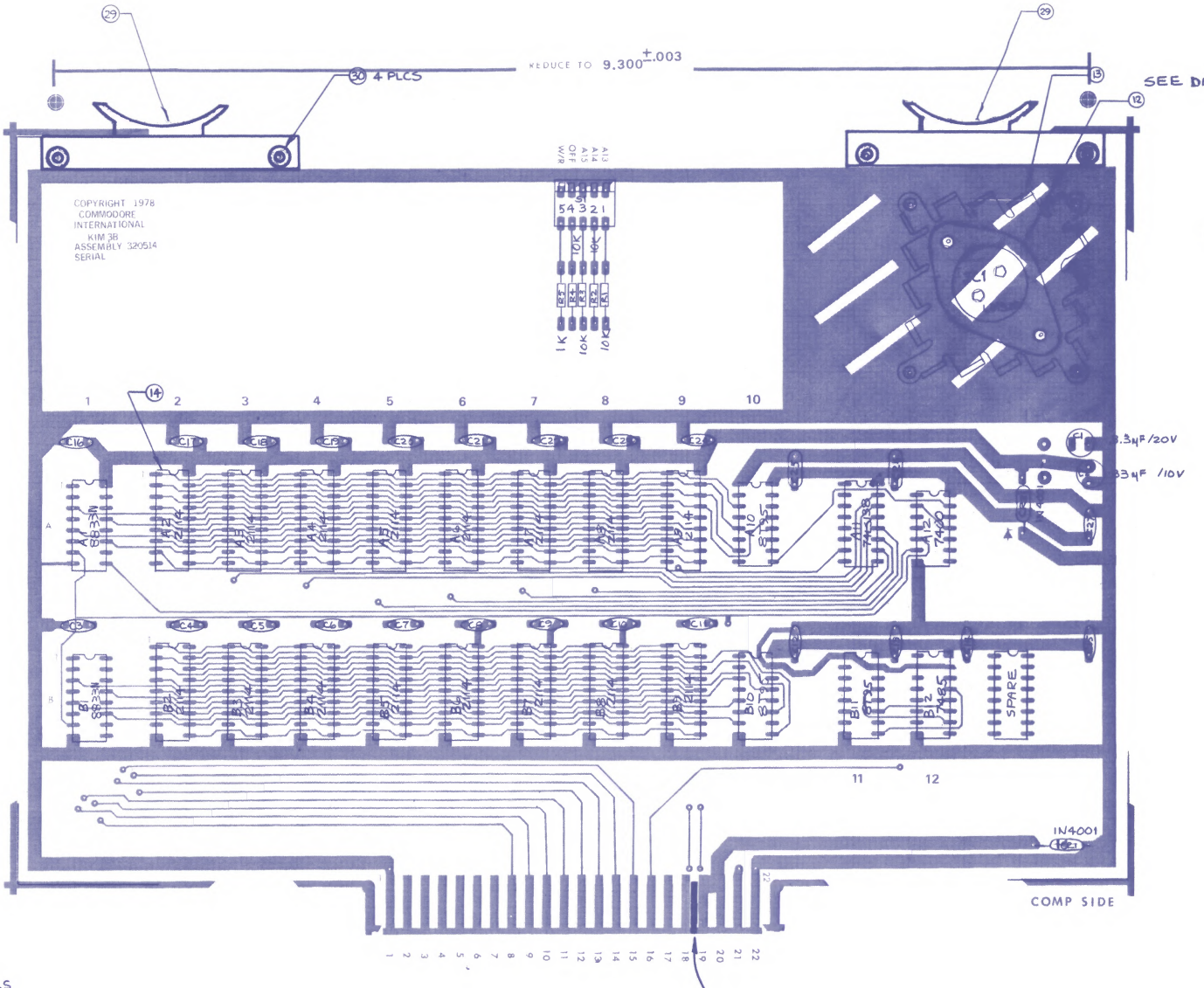

INPUT
BUFFERS
OUTPUT
BUFFERS
DATA
BUS
(8)
MEMORY
MATRIX
(8)
DATA
BUS
ADDRESS
DECODING
(16)
ADDRESS
BUS

lines (AB13 - 15) are presented to B12, a 4-bit comparator. The bit pattern on these lines is compared with the bit pattern generated by the three address switches. If the address switch settings and the high-order address lines match, pin 6 of B12 will go high. This signal is passed off the board as the board SELECTED line (BDSEL). When BDSEL goes high it disables U4 on KIM-1, preventing the memory circuits on KIM-1 from conflicting with addresses intended for the expansion memory board.

Address bus lines 10, 11 and 12 are decoded in A11 to provide eight output lines. Each of these eight lines is connected to two memory circuits to determine which memory will be active at any time. The memory circuits will not be activated unless the proper address configuration exists on address bus lines 13, 14 and 15. The comparison signal from pin 6 of B12 is also used in A12 to combine with the Ø2 and R/W signals from KIM-1 to control the input and output buffers (A1 and B1).

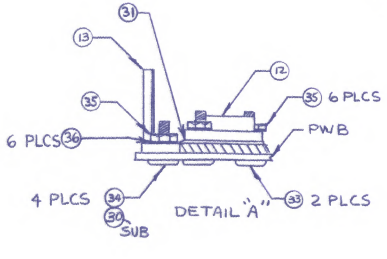
When clock phase 2 is present the data bus buffers will be enabled, allowing data to be fed into or out of the board, depending on the condition of the Read/Write line. All will enable the appropriate two memory circuits and the selected memory circuits will decode the ten least significant address bits to complete the Read/Write operation.

Also present is a conventional 3-terminal regulator, which takes the unregulated +8v supplied to the board and supply a regulated +5v for the circuitry.



NOTE. 1) ALL RESISTOR VALUES IN OHMS, TOLERANCE 5%, UNLESS OTERWISE NOTED.

2) ALL CAPACITORS ARE .1 MICRO FARAD, 10 VOLT UNLESS OTHERWISE NOTED.



ASSY, PCB, KIM 3B



MOS TECHNOLOGY, INC.

VALLEY FORGE CORPORATE CENTER

950 RITTENHOUSE ROAD, NORRISTOWN, PA. 19401

TEL: (215) 666-7950 TWX:510/660/4033

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